

AIM : Write a program to take 1D integer array convert it into Pandas and numpy

CODE:

```
import numpy as np
import pandas as pd

print("Enter integers separated by space:")
values = list(map(int, input().split()))

np_array = np.array(values)
pd_series = pd.Series(values)

print("\nNumPy Array:")
print(np_array)

print("\nPandas Series:")
print(pd_series)
```

OUTPUT:

```
PS D:\c programs> python -u "d:\c programs\tempCodeRunnerFile.python"
Enter integers separated by space:
1 3 5 6 7 8 9

NumPy Array:
[1 3 5 6 7 8 9]

Pandas Series:
0    1
1    3
2    5
3    6
4    7
5    8
6    9
dtype: int64
```

AIM : write a program to take 2D integer array and convert it into 1D numpy and Pandas**CODE:**

```
import numpy as np
import pandas as pd
rows = int(input("Enter number of rows: "))
cols = int(input("Enter number of columns: "))
matrix = []
print("Enter the matrix values row by row:")
for i in range(rows):
    row_vals = list(map(int, input().split()))
    matrix.append(row_vals)
np_2d = np.array(matrix)
numpy_1d = np_2d.flatten()
pandas_1d = pd.Series(numpy_1d)
print("NumPy 1D Array:")
print(numpy_1d)
print("\nPandas 1D Series:")
print(pandas_1d)
```

OUTPUT:

```
PS D:\c programs> python -u "d:\c programs\tempCodeRunnerFile.python"
Enter number of rows: 2
Enter number of columns: 2
Enter the matrix values row by row:
1 2
5 6
NumPy 1D Array:
[1 2 5 6]

Pandas 1D Series:
0    1
1    2
2    5
3    6
dtype: int64
```

AIM : Write a program to take 5 different coordinate value and find out the total error in y coordinate with respect to any line $y = mx+c$. And plot it.

CODE:

```
import numpy as np

import matplotlib.pyplot as plt

m = float(input("Enter value for m: "))
c = float(input("Enter value for c: "))

x_data = []
y_data = []

print("Enter coordinates for 5 points:")
for i in range(5):
    print("Point", i+1)
    x_val = float(input("Enter x: "))
    y_val = float(input("Enter y: "))
    x_data.append(x_val)
    y_data.append(y_val)

x = np.array(x_data)
y = np.array(y_data)

y_pred = m * x + c
errors = y - y_pred

sse = np.sum(errors ** 2)

print("\nComputed Values:")
print("X:", x)
print("Actual Y:", y)
print("Predicted Y:", y_pred)
print("Errors:", errors)
print("Total Squared Error:", sse)

plt.scatter(x, y, color='blue', label='Actual')

x_line = np.linspace(np.min(x), np.max(x), 10)
y_line = m * x_line + c

plt.plot(x_line, y_line, color='green', label='Equation')
```

```
for i in range(len(x)):
```

```
    plt.plot([x[i], x[i]], [y[i], y_pred[i]], 'r--')
```

```
plt.xlabel('X Axis')
```

```
plt.ylabel('Y Axis')
```

```
plt.legend()
```

```
plt.show()
```

INPUT:

```
PS D:\c programs> python -u "d:\c programs\tempCodeRunnerFile.python"
Enter value for m: 1
Enter value for c: 3
Enter coordinates for 5 points:
Point 1
Enter x: 4
Enter y: 5
Point 2
Enter x: 6
Enter y: 7
Point 3
Enter x: 2
Enter y: 4
Point 4
Enter x: 5
Enter y: 6
Point 5
Enter x: 8
Enter y: 9
```

OUTPUT:

```
Computed Values:
X: [4. 6. 2. 5. 8.]
Actual Y: [5. 7. 4. 6. 9.]
Predicted Y: [ 7.  9.  5.  8. 11.]
Errors: [-2. -2. -1. -2. -2.]
Total Squared Error: 17.0
■
```

PLOT: